

assignment2

February 4, 2025

1 Objectives

The objective of this assignment is to deepen the understanding of various **probability concepts** and their applications in real-world scenarios. Through a series of exercises, this assignment will help the student explore the following key concepts:

1. **Probability and Sample Spaces:** The assignment begins with the identification and listing of all possible outcomes for a series of random experiments. The student will develop the ability to define and interpret **sample spaces** for different events, such as rolling dice, flipping coins, and observing outcomes of random processes like blood type distributions.
2. **Disjoint and Independent Events:** The assignment requires the student to determine whether events are **disjoint** (mutually exclusive) or **independent** (where the occurrence of one event does not affect the occurrence of another). This will be done through two real-world scenarios where the relationships between events must be evaluated.
3. **Applying Probability Rules:** The student will apply basic probability rules, such as the **addition rule** and **multiplication rule**, to solve problems involving random events. Key applications include calculating the probability of various outcomes and understanding how events interact in complex scenarios.
4. **Conditional Probability:** A significant portion of the assignment focuses on **conditional probability**, where the student will calculate the probability of an event occurring given that another event has occurred. This section will include understanding how to calculate conditional probabilities using **two-way tables** and assess the independence of events.
5. **Normal Distribution:** The student will be introduced to **normal distributions** and will calculate probabilities associated with normal random variables. By applying the **z-score** formula, the student will analyze data and understand the concept of standard deviations in normal distributions.
6. **Sampling Distributions:** The assignment concludes with questions related to **sampling distributions**, helping the student differentiate between **population parameters** and **sample statistics**. The student will gain insights into the variability of statistics from random samples and learn how to interpret sampling distributions in the context of real-world data.

1.0.1 1.1) Toss a coin once and record whether you get heads (H) or tails (T).

- **Sample Space (S):** The possible outcomes are **Head (H)** or **Tail (T)**. Therefore, the sample space for this random experiment is: $S = \{H, T\}$

1.0.2 1.2) Toss a coin twice.

- **Sample Space (S):** When tossing a coin twice, we have two outcomes for each toss, leading to the following possible outcomes:
 - First toss is Head (H) and second toss is Head (H), resulting in HH.
 - First toss is Head (H) and second toss is Tail (T), resulting in HT.
 - First toss is Tail (T) and second toss is Head (H), resulting in TH.
 - First toss is Tail (T) and second toss is Tail (T), resulting in TT.

Therefore, the sample space for this random experiment is:

$$S = \{HH, HT, TH, TT\}$$

1.0.3 1.3) Toss a coin 3 times.

- **Sample Space (S):** For three tosses, each toss has two possible outcomes (Head or Tail), leading to a total of ($2^3 = 8$) possible outcomes. These outcomes are:

$$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$$

1.0.4 1.4) Select a person at random and check his or her blood type.

- **Sample Space (S):** The four common blood types are **A**, **B**, **AB**, and **O**. Therefore, the sample space is:

$$S = \{A, B, AB, O\}$$

1.0.5 1.5) Roll a fair die (a die having 6 faces of 1, 2, 3, 4, 5, 6 dots).

- **Sample Space (S):** When rolling a single die, there are six possible outcomes (one for each face of the die). The sample space is:

$$S = \{1, 2, 3, 4, 5, 6\}$$

1.0.6 1.6) A pair of fair dice is rolled, and the sum of the dots on the two faces that come up is recorded.

- **Sample Space (S):** When two dice are rolled, the sum of their dots can range from 2 ($1+1$) to 12 ($6+6$). The possible sums are:

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

1.0.7 1.7) A couple decides to have children until they have one boy and one girl, but they will not have more than three children.

- **Sample Space (S):** The couple has children until they have at least one boy and one girl, with no more than three children. The possible outcomes are:
 - First two children are Boy (B) and Girl (G), so the third child could be irrelevant: ($\{BG\}$)
 - First child is Boy (B), second is Girl (G) — and any order is possible.

Therefore, the sample space includes the following possibilities (since order matters and the child sequence stops after one of each gender):

$$S = \{BG, GB, BBG, GBB\}$$

1.0.8 Situation 1:

- **Context:** A music class consists of 4 male students and 4 female students. Two of the 8 students are chosen at random for a role-playing exercise.
- **Events:**
 - **Event A:** The first chosen student is male.
 - **Event B:** The second chosen student is female.

1.0.9 2.1) Are the two above events A and B disjoint or not disjoint? Explain why.

- **Definition of disjoint events:** Two events are **disjoint** (or mutually exclusive) if they cannot occur at the same time. In other words, the occurrence of one event means that the other cannot occur.

Explanation:

- Event A (the first student is male) and Event B (the second student is female) are **not disjoint**.
- This is because it is possible for **both events to happen** in the same trial. For example, if the first student chosen is male, and the second student chosen is female, both events A and B will occur.
- Therefore, these two events are **not disjoint** because they are not mutually exclusive.

Answer: **Not disjoint** because both events can occur together.

1.0.10 2.2) Are the two above events A and B independent or dependent? Explain why.

- **Definition of independent events:** Two events are **independent** if the occurrence of one event does not affect the occurrence of the other event. If the probability of event B occurring remains the same regardless of whether event A has occurred, then the events are independent.

Explanation:

- Event A (the first student is male) and Event B (the second student is female) are **dependent** events.
- This is because the outcome of the first selection (whether a male or female is chosen) affects the probability of the second selection.
- If the first student chosen is male, there will be fewer males remaining in the group, which affects the probability of the second student being female.
- Therefore, these events are **dependent** because the selection of one student impacts the selection of the other.

Answer: **Dependent** because the outcome of the first event affects the probability of the second event.

1.0.11 Situation 2:

- **Context:** Roughly 5% of Canadian males have some form of color blindness. Suppose that two Canadian males are selected at random for a medical study.
- **Events:**
 - **Event C:** The first male is color blind.
 - **Event D:** The second male is not color blind.

1.0.12 2.3) Are the two above events C and D disjoint or not disjoint? Explain why.

- **Definition of disjoint events:** Two events are **disjoint** (or mutually exclusive) if they cannot occur at the same time.

Explanation:

- Event C (the first male is color blind) and Event D (the second male is not color blind) are **not disjoint**.
- These events are not mutually exclusive because it is possible that the first male is color blind, and the second male is not color blind. Both events can occur at the same time.

Answer: **Not disjoint** because both events can occur simultaneously.

1.0.13 2.4) Are the two above events C and D independent or dependent? Explain why.

- **Definition of independent events:** Two events are **independent** if the occurrence of one does not affect the occurrence of the other. If the probability of event D occurring remains the same regardless of whether event C occurred, then the events are independent.

Explanation:

- Event C (the first male is color blind) and Event D (the second male is not color blind) are **independent** events.
- The reason is that the condition of the first male (whether or not he is color blind) does not affect the condition of the second male, since the probability of the second male being color blind or not is not influenced by the first male's status. In other words, the selection of the first male does not affect the probability of the second male being color blind or not.

Answer: **Independent** because the probability of one event occurring is not affected by the other.

Summary Question 2:

- **Situation 1 (Music class):**
 - Events **A** and **B** are **not disjoint** because both events can occur together (first male and second female).
 - Events **A** and **B** are **dependent** because the first selection affects the probability of the second selection.
- **Situation 2 (Color blindness in Canadian males):**
 - Events **C** and **D** are **not disjoint** because they can occur together (first color blind, second not color blind).

- Events **C** and **D** are **independent** because the condition of the first male does not affect the condition of the second male.

1.0.14 Given Probabilities:

- Blood type O: $P(O) = 0.44$
- Blood type A: $P(A) = 0.42$
- Blood type B: $P(B) = 0.10$
- Blood type AB: $P(AB) = 0.04$

1.0.15 3.1) What is the probability if a randomly chosen person in the USA does not have blood type AB?

We already know that $P(AB) = 0.04$. The complement of this event (not having blood type AB) is:

$$P(\text{Not AB}) = 1 - P(AB) = 1 - 0.04 = 0.96$$

Answer: 0.96

1.0.16 3.2) What is the probability that a randomly chosen person is a potential donor for a blood type B person?

A person who can donate to a person with blood type B must have the following blood types: - Type O (can donate to anyone, including B): $P(O) = 0.44$ - Type B (can donate to other type B and AB): $P(B) = 0.10$ - Type AB (can donate to type AB only, not to B): This will not be included.

Therefore, the total probability of being a donor for a blood type B person is:

$$P(\text{Donor for B}) = P(O) + P(B) = 0.44 + 0.10 = 0.54$$

Answer: 0.54

1.0.17 3.3) What is the probability that a randomly chosen person is a potential donor for a blood type AB person?

A person who can donate to a person with blood type AB must have the following blood types: - Type O (can donate to anyone, including AB): $P(O) = 0.44$ - Type A (can donate to type A or AB): $P(A) = 0.42$ - Type B (can donate to type B or AB): $P(B) = 0.10$ - Type AB (can donate to type AB only): $P(AB) = 0.04$

Therefore, the total probability of being a donor for a blood type AB person is:

$$P(\text{Donor for AB}) = P(O) + P(A) + P(B) + P(AB) = 0.44 + 0.42 + 0.10 + 0.04 = 1.0$$

So, the probability is 1 (which makes sense because everyone can donate to AB in some way).

Answer: 1.0

1.0.18 3.4) Two people are selected at random from all people in the United States. What is the probability that both have blood type B?

For two people to both have blood type B, we multiply the individual probabilities for both people having blood type B:

$$P(\text{Both B}) = P(B) \times P(B) = 0.10 \times 0.10 = 0.01$$

Answer: 0.01

1.0.19 3.5) Two people are selected simultaneously and at random from all people in the United States. What is the probability that both have blood type O?

For two people to both have blood type O, we multiply the individual probabilities for both people having blood type O:

$$P(\text{Both O}) = P(O) \times P(O) = 0.44 \times 0.44 = 0.1936$$

Answer: 0.1936

1.0.20 3.6) Two people are chosen simultaneously and at random. What is the probability that both have the same blood type?

The event of having the same blood type can happen in four ways: - Both have blood type O: $P(\text{Both O}) = P(O) \times P(O) = 0.44 \times 0.44 = 0.1936$ - Both have blood type A: $P(\text{Both A}) = P(A) \times P(A) = 0.42 \times 0.42 = 0.1764$ - Both have blood type B: $P(\text{Both B}) = P(B) \times P(B) = 0.10 \times 0.10 = 0.01$ - Both have blood type AB: $P(\text{Both AB}) = P(AB) \times P(AB) = 0.04 \times 0.04 = 0.0016$

To find the probability that both people have the same blood type, we sum these probabilities:

$$P(\text{Same blood type}) = P(\text{Both O}) + P(\text{Both A}) + P(\text{Both B}) + P(\text{Both AB})$$

$$P(\text{Same blood type}) = 0.1936 + 0.1764 + 0.01 + 0.0016 = 0.3816$$

Answer: 0.3816

	B	not B	Total
D	0.38	0.57	0.95
not D	0.02	0.03	0.05
Total	0.40	0.60	1.00

1.0.21 4.1) Using the above two-way probability table, find $P(D | B)$?

Conditional probability formula:

$$P(D|B) = P(D \text{ and } B) / P(B)$$

From the table: - $P(D \text{ and } B) = 0.38$ (probability that both the alarm in the dining room is triggered, and smoke is detected in the bedroom). - $P(B) = 0.4$ (probability that smoke was detected in the bedroom).

So:

$$P(DB) = 0.38 / 0.4 = \mathbf{0.95}$$

1.0.22 4.2) Using the above two-way probability table, find $P(D | \text{not } B)$

For the conditional probability $P(D | \text{not } B)$, we use the formula:

$$P(D | \text{not } B) = P(D \text{ and not } B) / P(\text{not } B)$$

From the table: - $P(D \text{ and not } B) = 0.57$ (probability that the dining room alarm triggers when no smoke is detected in the bedroom). - $P(\text{not } B) = 0.6$ (probability that smoke was not detected in the bedroom).

So:

$$P(D \mid \text{not } B) = 0.57 / 0.6 = \mathbf{0.95}$$

1.0.23 4.3) Use conditional probability, check whether the events D and B are independent or not

Two events D and B are independent if: $P(D|B) = P(D)$

From the table, $P(D) = 0.95$. We found: $P(D|B) = 0.95$ and $P(D|\text{not } B) = 0.95$

Since $P(D|B) = P(D)$ the occurrence of B does not affect the probability of D. Therefore D and B are **are independent events**.

1.0.24 Summary of Question 4 Answers:

- **4.1:** $P(D \mid B) = 0.95$
- **4.2:** $P(D|\text{not } B) = 0.95$
- **4.3:** The events D and B are **independent**.

1.0.25 5.1) Kyle earns \$23.5 an hour. How far is Kyle's hourly rate from the mean or what is his z-score?

The **z-score** formula is:

$$z = \frac{X - \mu}{\sigma}$$

Where: - X is the value of the data point (Kyle's hourly rate, \$23.5).

- μ is the mean (average salary, \$28).
- σ is the standard deviation (\$6).

Substitute the values into the formula:

$$z = \frac{23.5 - 28}{6} = -4.5/6 = \mathbf{-0.75}$$

So, **Kyle's z-score is -0.75**. This means his hourly rate is 0.75 standard deviations below the mean.

1.0.26 5.2) Sherry also works for "ABC Consulting Firm" and the z-score for her hourly salary is 2.6. What is Sherry's actual hourly salary?

We can rearrange the **z-score formula** to solve for Sherry's actual hourly salary (X):

$$X = \mu + z \times \sigma$$

Where: - μ is the mean (\$28).

- z is Sherry's z-score (2.6).

- is the standard deviation (\$6).

Substitute the values into the formula:

$$X = 28 + (2.6 \times 6) = 28 + 15.6 = 43.6$$

So, **Sherry's actual hourly salary is \$43.60.**

1.0.27 5.3) Tom also works for “ABC Consulting Firm” and the z-score for his hourly salary is -1.3. What is Tom's actual hourly salary?

Again, we can use the **z-score formula** to solve for Tom's actual hourly salary (X):

$$X = \mu + z \times \sigma$$

Where: μ is the mean (\$28).

- z is Tom's z-score (-1.3).
- σ is the standard deviation (\$6).

Substitute the values into the formula:

$$X = 28 + (-1.3 \times 6) = 28 - 7.8 = 20.2$$

So, **Tom's actual hourly salary is \$20.20.**

1.0.28 Summary of Answers:

- **5.1:** Kyle's z-score is **-0.75**.
- **5.2:** Sherry's actual hourly salary is **\$43.60**.
- **5.3:** Tom's actual hourly salary is **\$20.20**.

1.0.29 Situation 1: Human Pregnancy Length

- **Population Parameters:** These are the values that describe the entire population.
 - **Mean pregnancy length** = 266 days
 - **Standard deviation** = 16 days These values are given as the population parameters because they represent the entire population of human pregnancies.

Situation 1: Human Pregnancies

6.1) Population Parameters:

- **Mean (μ):** The true average length of human pregnancies is given as 266 days.
- **Standard Deviation (σ):** The true variability in pregnancy lengths is 16 days.

6.2) Sample Statistics:

- **Sample Mean ($\bar{x}$):** In the sample of 20 pregnant women, the average pregnancy length observed was 270 days.
- **Sample Standard Deviation (s):** The variability observed in the sample was 12 days.

Explanation:

The population parameters represent the true characteristics of the entire population of pregnancies, while the sample statistics are estimates based on the 20 women sampled. In this case, the sample

mean is slightly higher than the population mean, and the sample standard deviation is lower than the population standard deviation, indicating potential sampling variation.

1.0.30 *Situation 2: Part-Time University Students*

- **Population Parameters:** This is the value for the entire population of part-time university students in Canada.
 - **Proportion of female students** = 60% (0.60) This is the population parameter because it refers to the proportion of female students in the entire population of part-time students in Canada.

1.0.31 6.3) What is the population parameter from this situation?

- **Proportion (p):** It is stated that approximately 60% of all part-time university students in Canada are female. Thus, the population parameter is $p = 0.60$.

1.0.32 6.4) What is the sample statistic from this situation?

- **Sample Proportion ($\hat{p}$):** In a sample of 300 part-time students from ABC University, 155 were found to be female. This gives a sample proportion of:

$$155/300 = 0.5167.$$

Explanation:

The population parameter of 0.60 is the assumed true proportion of female part-time students across Canada. The sample statistic, approximately 0.5167, is the proportion calculated from the 300 students sampled at ABC University. The difference between the two values can be attributed to sampling variability or differences in the specific university population compared to the national level.